

A 100-neuron two-leg spinal CPG

Progress report on the neuron and synapse reduction study

1. Summary

The two-leg spinal CPG was reduced from the $k = 1$ baseline (**1,200 neurons, 118,208 synapses**) to **100 Izhikevich neurons and 2,710 synapses (2.3 %)**. Tested on 8 seeds never used for tuning, the reduced network passes every acceptance rule at all three gait speeds (350, 520 and 780 ms steps). It keeps flexor/extensor alternation, both legs, left–right anti-phase and 60 s stability, with STDP on. It follows the baseline more closely than the earlier 480-neuron control did: the flexor now switches fully off each cycle (trough 0.87 vs 3.78 at 520 ms).

Two changes made this possible at 100 neurons. The first is a minimum of 4 inputs per neuron, with the total weight conserved; without it every pathway has a single input and results depend on the seed. The second is a flexor rhythm-generator setting that changes with gait. The main remaining difference from the baseline is that at 520 ms the reduced network alternates more cleanly (-0.987 vs -0.955).

2. Goal and method

Starting point: the 480-neuron control ($k = 0.20$, $m = 0.40$, $\text{corr_F} \approx -0.94$). Goal: 100 neurons in total (50 per leg; relay parrots not counted), matching the $k = 1$ baseline over 60 s, with STDP on or off.

- **Model:** `cpg_small.py`, a copy of the original model with per-population sizes and extra switches. The originals were never edited.
- **Measurement:** 8 seeds, 1 thread, 60 s, first 10 s discarded. Two-pass readout gate per leg: pass A uses the stock threshold, pass B the threshold each leg's own RG-F band prescribes. Only pass B is reported.
- **Main metric:** corr_F = correlation of extensor and flexor force (-1 = perfect alternation). corr_RG is the same for the rhythm-generator rates and does not depend on the gate.
- **Pass rules (A1–A8), judged against the baseline at the same gait:** corr_F within ± 0.02 (the band is 0.02 wider on the more-alternating side for STDP-on cells); flexor trough ≤ 4 ; seed SD ≤ 0.03 ; force not ahead of rhythm; stable across run thirds; right leg in band; L–R phase 0.5 ± 0.05 ; rhythm no more than 0.02 weaker than baseline.
- **Evidence:** only runs on the Mac with NEST 3.9.0 count. The final recipe was confirmed on held-out seeds (13579...83579).

3. What was done

Step	Change tested	Outcome
E0	Re-run the $k = 1$ baseline and the 480 control at 60 s	Both reproduce. Found that a 20 ms trace/muscle step (the old default) breaks the model; 100 ms is required
E1	240 neurons; fixed in-degree vs random wiring	Fixed in-degree passes (-0.955 ± 0.010); random wiring fails (seed SD 0.11)
E2	Ladder 180 $\rightarrow$ 140 $\rightarrow$ 100	140 passes; 100 fails (SD 0.046) — every pathway has only 1 input per neuron
E3	Minimum 3 inputs per neuron + $E \rightarrow \text{InE} \rightarrow F$ inhibition $\times 1.5$	100 neurons pass at 520 ms (-0.970 ± 0.020)
E6	STDP on (λ 1e-5 $\rightarrow$ 1e-6)	1e-6 settles in 60 s; passes
E8	All gaits, both legs, L–R phase	520 ms passes everything; 350 ms seed SD 0.038; 780 ms rhythm too weak
E5	Make RG-F burst (Izhikevich $c = -50$, adaptation $d \leq 2$)	Fixes 780 ms; at 520 ms the right leg alternates more than the baseline

Step	Change tested	Outcome
E5b	Minimum 4 inputs + RG-F setting chosen per gait	Every gait passes (350 ms SD 0.038 → 0.018)
E8f	Final recipe on 8 held-out seeds, STDP on and off	Confirmed: passes all rules at all gaits (STDP on)
E3g, E8g	Weaker E→InE→F inhibition ($\times 0.67$ – $\times 1.0$) to close the 520 ms gap	Closer mean, larger spread; $\times 1.0$ fails 520/780 on held-out seeds → not adopted

4. Final result (held-out seeds)

Gait	Network	corr_F left	corr_F right	corr_RG	Flexor trough
350 ms	k = 1 baseline (1,200 n.)	-0.918 ± 0.014	-0.917	-0.895	2.52
	100 neurons	-0.927 ± 0.019	-0.922	-0.921	2.10
520 ms	k = 1 baseline	-0.955 ± 0.013	-0.942	-0.938	0.83
	100 neurons	-0.987 ± 0.005	-0.982	-0.958	0.87
	480-neuron control	-0.943 ± 0.024	-0.907	-0.948	3.78
780 ms	k = 1 baseline	-0.994 ± 0.003	-0.993	-0.930	0.25
	100 neurons	-0.992 ± 0.003	-0.993	-0.921	0.28

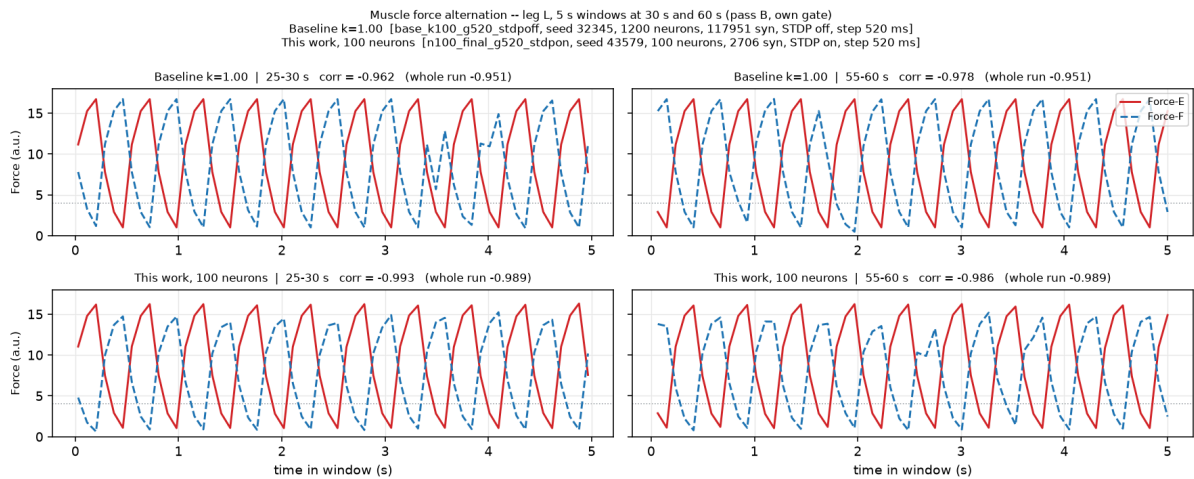
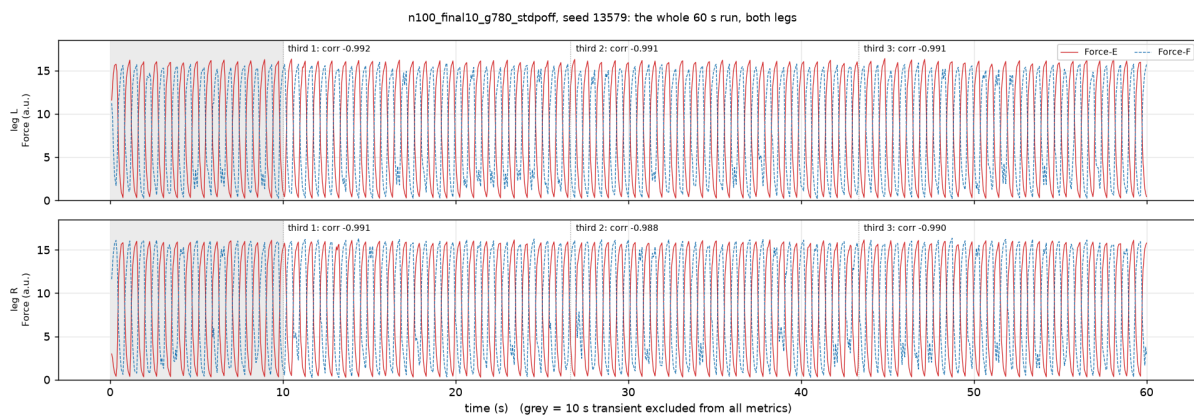


Figure 1. Muscle force at 520 ms, left leg, 5 s windows at 30 s and 60 s. Top: $k = 1$ baseline. Bottom: 100-neuron network (held-out seed 43579).



5. Final recipe

- **Sizes per leg:** RG-E 10, RG-F 10, InE 6, InF 8, Ia-interneurons 4 + 4, MN-E 4, MN-F 4 (50 per leg); connectivity $k = 0.20$; 40 relays.

- **Wiring:** fixed in-degree with at least 4 inputs per neuron, weights scaled to keep the total input unchanged (`--indeg-min 4 --indeg-min-conserve`).
- **Inhibition:** $E \rightarrow \text{InE} \rightarrow F$ pathway $\times 1.5$ over the $k = 0.20$ recipe (`--inh-comp 5.625`).
- **Flexor rhythm generator by gait:** $c -55$, $d 4$ at 350–520 ms; $c -50$, $d 2$ at 780 ms (`--rgf-c`, `--rgf-d`).
- **STDP:** on, $\lambda = 1e-6$ (the result is the same with STDP off, except at 520 ms, see limitations).

6. Key findings

1. The trace/muscle update step must be 100 ms; the old 20 ms default gives a different model (rhythm -0.63 vs -0.95).
2. Each leg needs its own flexor gate: the legs receive opposite RG-F bias currents.
3. At small sizes, random wiring fails through seed spread; fixed in-degree fixes it.
4. At 100 neurons, raising connectivity k cannot help: every pathway rounds to 1 input. A minimum in-degree (3–4) is the lever.
5. $E \rightarrow \text{InE} \rightarrow F$ inhibition is the sensitive hand-tuning knob. Less inhibition brings the mean closer to the baseline but increases seed spread.
6. Reduced networks never reproduce the baseline's 0–500 Hz RG-F bursting. Lower RG-F adaptation is needed for the slow gait.
7. Choosing from many variants on the same seeds is optimistic; the held-out check changed one decision ($\times 1.0$ inhibition).

7. Limitations

- At 520 ms the reduced network alternates more cleanly than the baseline (-0.987 vs -0.955). It passes only through the wider STDP-on band; with STDP off it misses by about 0.02–0.03.
- STDP has no measurable effect in the final network (differences ≤ 0.005).
- RG-F fires at ~ 80 –230 Hz without the baseline's deep bursts; RG-E carries most of the rhythm.
- One parameter (RG-F adaptation) must change with gait; the baseline needs none.
- Results are bit-reproducible on one machine and NEST 3.9.0, not across NEST versions.